

Regression Assignment

Step 1: Problem Statement

The client wants to **predict insurance charges** based on several parameters (age, sex, BMI, children, smoker and Charges).

Stage1: Machine Learning

Stage2: Supervised Learning

Stage3: Regression

Step 2: Dataset Information

- **Rows:** 1338
- **Columns:** 6
 - age (Input: numeric)
 - sex (Input: categorical)
 - bmi (Input: numeric)
 - children (Input: numeric)
 - smoker (Input: categorical)
 - charges (Output: numeric)

Step 3: Pre-processing

- Convert categorical variables (sex, smoker) into numeric using **One-Hot Encoding**.
- Check for missing values (none in this dataset).
- Standard Scalar features if needed.

Step 4: Model Development

We will try multiple regression models:

Models Tested

1. Multiple Linear Regression

- **R2 score :** 0.7894

2. Support Vector Machine

- **R2 score :** 0.6340 (kernel = "linear", C = 1000, *epsilon* = 0.1)

Support Vector Machine (Without StandardScaler)					
Hyper Tuning parameters					
		Kernel			
<i>C</i>	<i>epsilon</i>	Linear (R ² values)	Poly (R ² values)	Rbf (R ² values)	Sigmoid (R ² values)
0.1	0.1	-0.1220	-0.0862	-0.0895	-0.0897
100	0.1	0.5432	-0.0997	-0.1248	-0.1181
1	1	-0.1115	-0.0642	-0.0884	-0.0899
1000	.1	0.6340			

3. Decision Tree Regressor

- **R2 Score** : 0.6952 (criterion = "squared_error", splitter = "best", max_features = None)

4. Random Forest Regressor

- **R2 Score** : 0.8487 (n_estimators = 50, max_features= None , criterion = "poisson" ,random_state = 0)

Step 5: Results (R² Scores)

Model	R ² Score
Multiple Linear Regression	0.7894
Support Vector Machine	0.6340
Decision Tree	0.6952
Random Forest	0.8487

Step 6: Final Model Selection

- **Chosen Model:** Random Forest Regressor
- **Reason:** It gives the **highest R² score** of 0.8487. Random Forest is the most reliable and accurate model for predicting insurance charges in this dataset comparing to other models.

